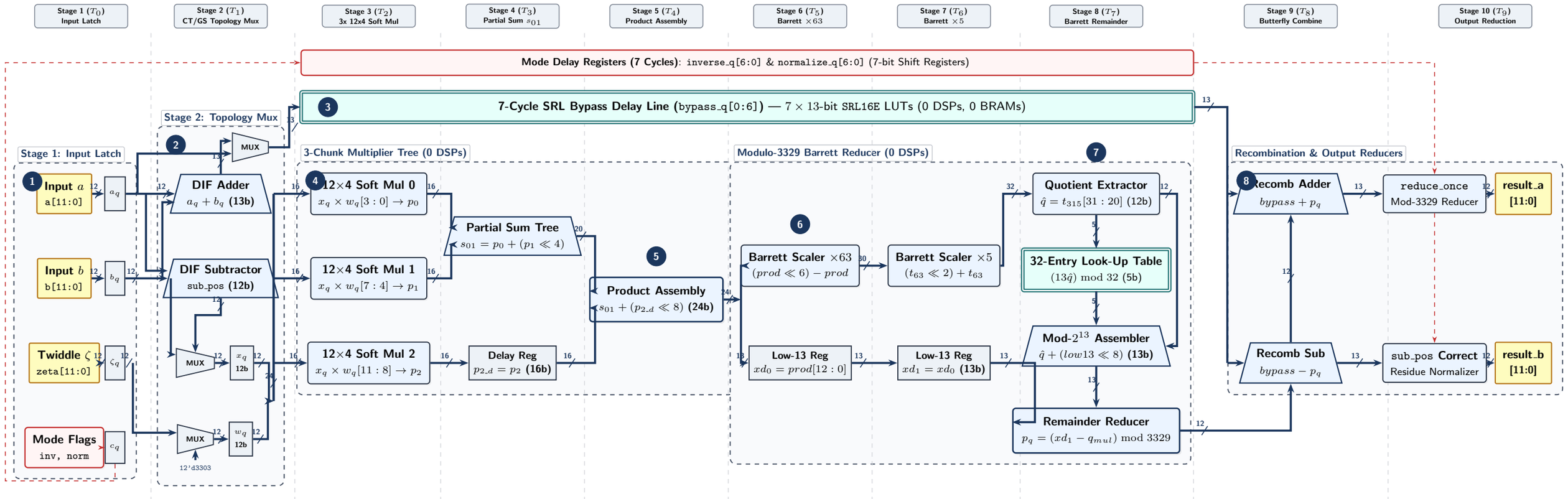
ML-KEM HPKA: 10-Cycle Zero-DSP Pipelined Butterfly Unit & Barrett Reducer

Subsystem 05 Microarchitecture — RTL Modules: `mlkem_butterfly.sv`, `mlkem_twiddle.svh` — Kintex UltraScale+ 450 MHz



ID	Component / Signal	Width / Format	Description
1	Input Operands & Latch	$3 \times 12\text{b} + 2 \times 1\text{b}$	Captures operands a, b, twiddle factor ζ, and control flags (`inv, norm`) into pipeline registers at clock edge T_0.
2	CT/GS Topology Selector	13b Add / 12b Sub	Muxes forward CT ($A \pm \omega B$) and inverse GS ($A + B, \omega(A - B)$) paths, computing sum $a + b$ and difference `sub_pos`.
3	7-Cycle SRL Bypass Line	13-bit × 7 Deep	Implements delay line for upper butterfly operand using AMD UltraScale `SRL16E` primitives, eliminating 91 flip-flops.
4	3-Chunk Soft Multiplier	3×16-bit Partial	Decomposes 12b × 12b multiplication into three parallel 12b × 4b soft multipliers (p_0, p_1, p_2) in CLB logic at 450 MHz.
5	Product Assembly Adder	20b → 24b Product	Two-stage carry-lookahead adder tree assembling partial sum $s_{01} = p_0 + (p_1 \ll 4)$ and full 24-bit product $prod = x \cdot w$.
6	Multiplier-Free Barrett Scaler	30b → 32b Shift-Add	Evaluates $prod \times 315$ via two shift-add stages (×63 and ×5), where $315 = \lfloor 2^{20}/3329 \rfloor$, avoiding DSP blocks.
7	Barrett Mod-3329 Remainder Unit	32×5b ROM / 12b	Exploits $q = 13 \cdot 2^8 + 1$; indexes 32-entry LUT with $\hat{q}[4:0]$ to reconstruct $q_{\text{mul}} \bmod 2^{13}$, emitting residue $p_q \in [0, 3328]$.
8	Butterfly Recombination & Reducers	$2 \times 13\text{b} \rightarrow 2 \times 12\text{b}$	Recombines bypass operand with product residue p_q, followed by `reduce_once` subtraction and normalizer correction for $result_a, result_b$.